Raj Kumar

Lead Backend Engineer | Python, TypeScript & AI Applications

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PROFESSIONAL SUMMARY

Backend engineer and systems architect with 8+ years of experience building APIs, enterprise integrations and data-driven applications. Hands-on expertise in Python, FastAPI, TypeScript, NestJS, PostgreSQL, MongoDB and asynchronous workers. Delivered EDI/AS2 and payroll integrations, real-time services, and provider-independent AI workflows. Independently launched mobile apps and browser tools through CompileCraft.

TECHNICAL SKILLS

Backend: Python, FastAPI, AsyncIO, TypeScript, JavaScript, Node.js, NestJS, Express, PHP, REST APIs, WebSockets, Socket.IO

Data & workers: PostgreSQL, SQLAlchemy, MongoDB, Mongoose, MySQL, Redis, BullMQ, ARQ, SQL, schema design, caching, idempotent processing

AI & integrations: Gemini / Vertex AI, ComfyUI, OpenAI SDK, OpenCV, X12 EDI, AS2, ERP adapters, Xero payroll, Stripe, Shopify webhooks

Delivery & security: Docker, Git, Linux, AWS S3 / SQS / Secrets Manager, JWT, OAuth2, audit logging, pytest, Jest, Supertest

PROFESSIONAL EXPERIENCE

Lead Backend Engineer & Systems Architect | Luminoguru Pvt Ltd

2020 - PRESENT | Mohali, India

- Built multi-tenant X12 EDI middleware with signed AS2 transport and ERP adapters.
- Developed TGA-Mesh workforce services with queued Xero payroll sync and real-time messaging.
- Designed provider-independent AI generation orchestration and image-processing services.

Backend & Web Developer | Quadrish Innovations

2019 - 2020 | Chandigarh, India

- Built and maintained REST APIs using PHP frameworks and MySQL.
- Integrated Node.js and MongoDB modules for asynchronous data workflows.

Software & Mobile Engineer (Trainee to Developer) | Tuespot Pvt Ltd

2017 - 2019 | Mohali, India

- Developed Java/XML Android features with SQLite storage and REST integration.
- Contributed PHP/SQL backend features and database schema design.

Raj Kumar | Selected Projects

SELECTED ENGINEERING PROJECTS

EDI Middleware & AS2 Gateway | Professional project

Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, ARQ, Docker

- Implemented signed and encrypted AS2 transport with synchronous and asynchronous MDN receipts.
- Added idempotent document ingestion, tenant configuration and append-only audit records.

TGA-Mesh | Professional project

TypeScript, NestJS, MongoDB, Redis, BullMQ, Xero, Socket.IO

- Implemented shift and leave rules with worker-based timesheet aggregation.
- Integrated queued Xero payroll sync, encrypted credentials and retry handling.

FashionTryOn | In development / production proof of concept

Python, FastAPI, NestJS, Redis, Shopify, Gemini, ComfyUI

- Unified Gemini, ComfyUI and Decart/FASHN generation pipelines behind a provider adapter interface.
- Designed queued generation with callbacks so GPU workloads do not block storefront requests.

StenGen | Professional project

TypeScript, NestJS, Python, OpenCV, Vector processing

- Separated ingestion, validation and user sessions from image-processing workloads.
- Combined OpenCV image quantization and vector processing for stencil-oriented output.

INDEPENDENT PRODUCT LAUNCHES - COMPILECRAFT

Creditora: Android finance app for khata, repayments, EMI tracking and shared expenses.

https://play.google.com/store/apps/details?id=in.compilecraft.creditora

HandBrowse: Chrome extension for camera-based gesture browsing with on-device processing.

https://handbrowse.compilecraft.in

Chess Royal: Android chess app with an AI coach, computer opponent and local play.

https://play.google.com/store/apps/details?id=in.compilecraft.chessroyal

Face Dodge: Android survival game with camera-based face movement controls.

https://play.google.com/store/apps/details?id=in.compilecraft.facedodge

EDUCATION

Master of Computer Applications (MCA) | MRSPTU, Punjab | 2016 - 2018

Bachelor of Computer Applications (BCA) | Panjab University, Chandigarh | 2012 - 2015